Grad Café Admissions Data Preparation & Statistical Analysis

Module 8 Analytical Report — Empirical Findings, Limitations & Hypotheses

Course: Johns Hopkins University — Principles of Software Development / Software Concepts (EN.605.601)
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Target Metric / Question	Calculated Empirical Value	Sample Size & Sub-Cohort	Significance / Status
Q1: Accepted PhD Average GRE	**162.36 pts** (Mean)	n = 2,495 accepted PhD applicants	Valid non-zero GRE scores
Q2: Accepted Master's Average GPA	**3.70 / 4.00** (Mean: 3.6997)	n = 4,531 accepted Master's applicants	Standard 4.0 scale (has_valid_gpa=1)
Q4: GRE Quant vs. GRE Verbal (Pearson)	**r = 0.0194, p = 0.1282**	n = 6,126 applicants with both scores	Not Statistically Significant (p > 0.05)
Q4: GPA vs. GRE Quant (Pearson / Spearman)	**r = 0.0265 (p = 0.0229)** rho = 0.0266 (p = 0.0223)	n = 7,390 valid paired records	Statistically Significant, Negligible Magnitude
Q5: GPA Accepted vs. Rejected (Welch t-test)	**t = -1.0357, p = 0.3003** (Mann-Whitney U: p = 0.3008)	Accepted: 3.6985 (n=8,290) Rejected: 3.7013 (n=8,158)	No Significant Difference (p > 0.05)
Q6: Degree vs. US/International (Chi-Square)	**chi2 = 0.0670, p = 0.7957** (df = 1)	N = 30,500 total applicants	Independence Supported (p >> 0.05)
Q6: Degree vs. Outcome (Chi-Square)	**chi2 = 6.2688, p = 0.0435** (df = 2)	N = 30,500 total applicants	Statistically Significant Association (p < 0.05)

1. Baseline Cohort Metrics: PhD GRE and Master's GPA

Question 1 (PhD Accepted GRE): Among applicants who were accepted into a doctoral program (Degree = 'PhD', outcome = 'Accepted') and reported a non-zero Quantitative GRE score, the average score is **162.36 points** (n = 2,495, SD = 4.62, median = 162.0). Scores span from 155 to 170 points, indicating that successful doctoral candidates generally perform at or above the 75th percentile of the ETS GRE Quantitative distribution.

Question 2 (Master's Accepted GPA): Among applicants who were accepted into a Master's program (Degree = "Master's", outcome = 'Accepted') with an undergraduate GPA evaluated on the standard 4.0 scale (0.0 <= GPA <= 4.0), the average GPA is **3.70** (specifically 3.6997, n = 4,531, SD = 0.174, median = 3.70). This demonstrates that Master's admits reflect strong prior academic records with minimal dispersion around the mean.

2. Shortcomings of Inferring Decision Dates from Status Text & Term Values

Question 3: Inferring admissions decision dates by concatenating an unstructured text date (e.g., 'Accepted on 15 Feb') with the calendar year parsed from the application term (e.g., 'Fall 2024' -> 2024) introduces three critical analytical shortcomings:

- **Academic Cycle vs. Calendar Year Misalignment:** Graduate admissions cycles routinely span calendar years. For instance, an applicant applying for 'Fall 2024' may receive an interview invitation or early acceptance notification in November or December 2023. Blindly assigning the term year (2024) erroneously pushes late-fall notifications forward by an entire year (to

November/December 2024), falsifying decision latency and timeline sequences. Conversely, 'Spring' terms frequently issue offers during late summer of the prior year.

- **Decoupling of Event Date from Self-Reporting Date:** On crowdsourced platforms like Grad Café, applicants frequently post retrospective entries weeks or months after receiving their official letter, or submit historical admissions data during subsequent application cycles. When a user posts on March 31, 2024 regarding an acceptance received in a prior cycle, associating the submission with the current cycle's term year distorts multi-year trend analysis.
- **Lack of Timestamp Precision & Typographical Fragility:** Free-text status fields lack timezone, hour, and standardization. Entries such as 'Accepted on 2/15', 'Accepted late Feb', or 'Accepted mid-March' resist automated parsing or default to invalid dates. Furthermore, records marked solely as 'Wait listed' or 'Interviewed' contain no decision date whatsoever, creating systematic missingness (10.23% in this dataset) that cannot be imputed without introducing substantial bias.

3. Correlation Analysis: Relationship Between GPA and GRE Measures

Question 4: We examined both parametric (Pearson *r*) and rank-based non-parametric (Spearman *rho*) correlation coefficients across all valid numeric dimensions:

• **GRE Quantitative vs. GRE Verbal:** Evaluated on n = 6,126 applicants possessing both valid scores, the Pearson correlation is **r = 0.0194 (p = 0.1282)**. Because the p-value exceeds the standard significance threshold (alpha = 0.05), we fail to reject the null hypothesis of zero correlation. In this dataset, Quantitative and Verbal standardized test capabilities are statistically independent.

• **Undergraduate GPA vs. GRE Quantitative:** Evaluated on n = 7,390 valid paired applicants on the standard 4.0 scale, the Pearson correlation is **r = 0.0265 (p = 0.0229)** and the Spearman rank correlation is **rho = 0.0266 (p = 0.0223)**. While the very large sample size renders this relationship statistically significant at alpha = 0.05, the effect size is practically negligible (accounting for less than 0.07% of shared variance, R^2 = 0.0007).

• **Analytical Interpretation:** These findings demonstrate that undergraduate GPA and standardized GRE scores measure fundamentally orthogonal dimensions of applicant ability (cumulative multi-year course performance vs. timed test-taking aptitude). For admissions committees, neither metric serves as an effective proxy for the other, justifying holistic, multi-criteria evaluation pipelines.

4. Hypothesis Testing: GPA Differences Between Accepted and Rejected Applicants

Question 5: To determine whether academic undergraduate performance differentiates admissions outcomes, we conducted two-sample hypothesis testing comparing valid GPAs of accepted applicants (n = 8,290) against rejected applicants (n = 8,158):

• **Descriptive Parameters:** Accepted cohort mean GPA = **3.6985** (SD = 0.174, median = 3.70). Rejected cohort mean GPA = **3.7013** (SD = 0.174, median = 3.70). The difference in sample means is a minuscule -0.0028 GPA points.

• **Welch's Two-Sample t-Test:** Accounting for potentially unequal population variances, the Welch t-test yielded a test statistic of **t = -1.0357** with a two-tailed p-value of **p = 0.3003**.

• **Mann-Whitney U Non-Parametric Test:** To guard against non-normality and bounded distributions, the Mann-Whitney U test yielded **U = 33,499,895.5** with a corresponding two-tailed p-value of **p = 0.3008**.

• **Substantive Conclusion:** Both parametric and non-parametric tests fail to reject the null hypothesis at any conventional significance level (alpha = 0.05, 0.01). Rejected applicants exhibit virtually identical undergraduate GPAs to accepted applicants. This counter-intuitive finding indicates that in competitive graduate admissions, undergraduate GPA acts as a threshold qualifier rather than a decisive differentiator. Admissions outcomes are governed primarily by non-GPA criteria—such as research alignment, faculty sponsorship, letters of recommendation, statement of purpose quality, and programmatic cohort capacity.

5. Categorical Association Analysis: Chi-Square Contingency Tests

Question 6: We evaluated independence between categorical variables using Pearson's Chi-Square Test (scipy.stats.chi2_contingency):

• **Degree Level vs. Residency Status (Master's/PhD vs. American/International):**

- Contingency Table: Master's [American: 8,054, International: 8,787] | PhD [American: 6,511, International: 7,148].
- Test Statistic: **chi2 = 0.0670, degrees of freedom = 1, p = 0.7957**.
- Interpretation: The proportion of international applicants is nearly identical across Master's (52.18%) and PhD (52.33%) cohorts. We firmly retain the null hypothesis of independence; degree level does not systematically influence an applicant's domestic vs. international residency distribution.

• **Degree Level vs. Admissions Outcome (Master's/PhD vs. Accepted/Rejected/Waitlisted):**

- Contingency Table: Master's [Accepted: 7,514, Rejected: 7,632, Waitlisted: 1,695] | PhD [Accepted: 6,237, Rejected: 5,996, Waitlisted: 1,426].
- Test Statistic: **chi2 = 6.2688, degrees of freedom = 2, p = 0.0435**.
- Interpretation: Because $p < 0.05$, we observe a statistically significant association between degree level and admissions outcome. Doctoral applicants experienced slightly higher acceptance rates (45.66% vs. 44.59%) and lower rejection rates (43.90% vs. 45.30%) than Master's applicants, reflecting distinct pool dynamics and funding models across graduate degree types.

6. Residual Data Quality Limitations in the Cleaned Dataset

Question 7: Despite rigorous standardization, whitespace collapsing, datetime parsing, and numeric validation, four critical data quality limitations persist within the cleaned dataset:

- **Self-Reporting and Survivorship Selection Bias:** Grad Café data is voluntarily crowdsourced. Applicants with high test scores (mean GRE = 162.5) and successful admissions outcomes are heavily overrepresented. Rejections, lower-tier institutions, and applicants with sub-3.40 GPAs are systematically underreported, skewing descriptive baselines toward positive outliers.
- **Pervasive Standardized Score Missingness:** Over 59.94% of records lack Quantitative GRE scores, and 79.91% lack Verbal and Analytical Writing scores. This extreme missingness reflects widespread institutional 'GRE optional' policies and selective reporting. Listwise deletion reduces usable analytical subsets by up to 80%, threatening sample representativeness.
- **GPA Clumping and Non-Standard Foreign Scales:** While the dataset was filtered to standard 4.0 scales, international applicants frequently self-convert from 10-point, 20-point, or percentage scales using unverified third-party formulas. Furthermore, anomalous entries such as GPA 4.61 (weighted high school scales or foreign conversion errors) create artificial outliers that distort statistical variance.
- **Uncontrolled Institutional & Program Taxonomy:** Free-text fields exhibit severe lexical variation (e.g., 'McG' vs. 'McGill University', 'UCB' vs. 'UC Berkeley'). Without standard CIP (Classification of Instructional Programs) or IPEDS institutional identifiers, cross-program generalizations remain susceptible to categorization errors.

Deliverable & Cloud Verification Confirmation:

All statistical values, hypothesis tests, and summary tables were programmatically generated from the validated dataset within **module_8.ipynb**. Data ingestion was executed from Amazon S3 bucket grad-cafe-nf-594366804861, and the final standardized output cleaned_gradcafe.json (30,500 records) was written back to S3. In compliance with cloud cost management guidelines, the Amazon SageMaker notebook instance was safely stopped.